

Andrew W. Taylor — B.Sc.

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EDUCATION

B.Sc., Biology

Trinity Western University

2020 – 2024

- Coursework: Advanced Molecular Biology; Molecular Genetics; Biochemistry; Medical Microbiology; Cellular Biology; Organic Chemistry; Evolutionary Theory; Ecology; Vertebrate Physiology.

Computer Science Coursework

University of British Columbia

2024 – 2025

- Coursework: Data Structures and Algorithms; Computer Systems; Software Construction; Models of Computation; Probability; Cognitive Systems Design.

EXPERIENCE

Research Assistant, R&D;

Vitacore Industries Inc.

Feb 2025 – Present

- Principal implementer across Vitacore's R&D; engineering portfolio: PCR/qPCR diagnostics, bioprocessing automation, print-farm infrastructure for medical-device fabrication, embedded hardware, and internal production tooling. Typically owns projects from first prototype through wet-lab validation and deployment.
- Convolutional PCR diagnostics: assay development with short primers and universal-tail chemistries; wet-lab validation (DNA extraction, gel electrophoresis, melt-curve/Ct analysis); bioinformatics for primer candidate evaluation and AI-assisted interpretation of multiplexed amplification signatures; custom PCB thermal cyclers for rapid nucleic acid cycling.
- Algae cultivation and carbon isotope separation: designed and built large-scale automated microalgae cultivation apparatus (multi-board ESP32 motor control, custom PCBs, FFT-based RPM stabilization); cloud-deployed workflow automation; culture, fermentation, and Drosophila feeding trials on fermented algal diets.
- Print farm automation: Mac mini server for 25+ industrial 3D printers with live feeds, runtime process injection for print control, worker-pool RAM management, and STL-to-slice pipelines for custom CPAP interface manufacturing.
- Internal tooling: procurement workflows, progressive web app notifications, engineering task tracking, and Chrome extensions for vendor-page order prefilling.
- Additional R&D:: STL-to-vasculature modeling for tissue scaffolds; fluorescent oxygen sensor firmware; NIR plastic-identification electronics; laboratory chemistry software (RDKit structure tools, Roche LightCycler analysis dashboards).

R&D; Intern

Vitacore Industries Inc.

May – Aug 2024

- Developed 3D vasculature generation algorithms (Unity/C# and Python) for biomedical scaffold design; built embedded motor-control systems and sensing prototypes across molecular biology and engineering projects.

Lab Instructor, Chemistry

Trinity Western University

2022 – May 2024

- Instructed undergraduate laboratory sections in qualitative and quantitative analysis, including titrations, gravimetric methods, and instrumentation; received excellent evaluations.

Laboratory Assistant, Chemistry

Trinity Western University

2021 – May 2024

- Prepared chemicals and equipment for chemistry labs; maintained safety protocols and waste disposal compliance; facilitated Science in the Valley outreach labs for visiting high school classes.

TECHNICAL SKILLS

Languages: Python; JavaScript/TypeScript; C++; C; C#; HTML/CSS

Science & ML: PCR/qPCR; melt-curve analysis; PyTorch; bioinformatics; scientific computing; wet-lab protocols

Software: FastAPI; Flask; React/Next.js; Node.js; Docker; Git; Playwright

Hardware: ESP32; PIC16; PCB design (KiCad/EasyEDA); PWM/I2C; 3D printing automation

CERTIFICATIONS

- WHMIS (Workplace Hazardous Materials Information System)
- Standard First Aid CPR/AED Level C